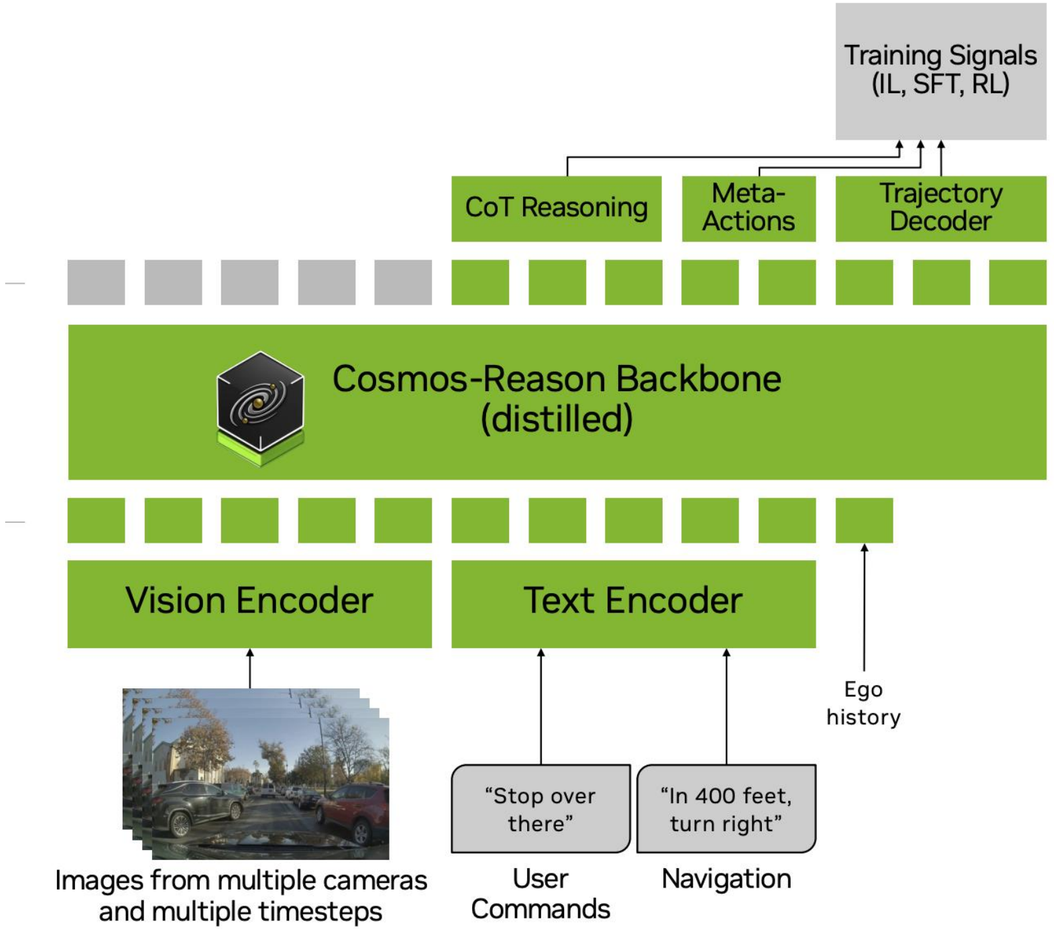
Alpamayo-R1

논문리뷰

AVLAB (Autonomous Vehicle Lab)

양경진 | 2026. 09. 22.

1. Architecture



Modular VLA

- input: Vision Encoder(DiNOv3), Ego history, Text Encoder(Option)
- Vison Encoder: Single Camera(논문), Triplain(2025), Flex(2025)

CoC Dataset

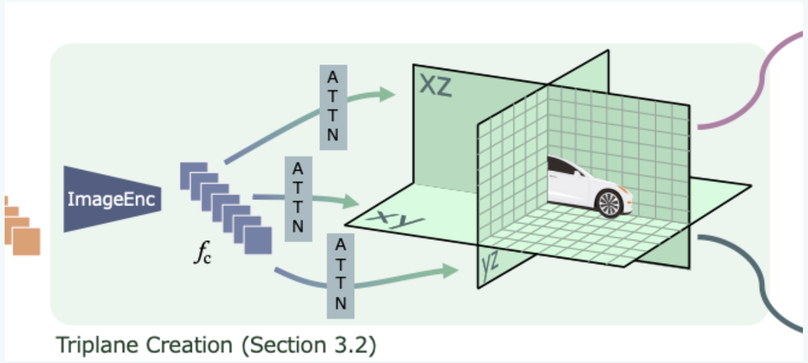
- 70만개의 Video clips
- Structured CoC(Causal Factors, Driving Decision, CoC Trace)

Multi step training

- SFT(Supervised Fine Tuning) + RL(Reinforcement Learning)

2. Vision Encoder

Triplane(2025, nvidia)



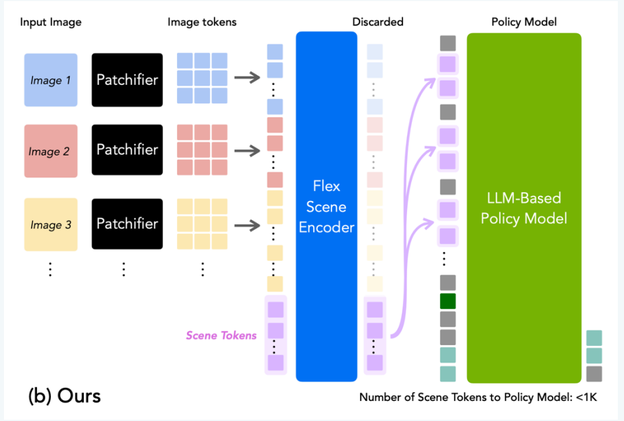
N cams (한 timestep)

ImageEnc: DINOv2-small (22M)

3D query grid 96×96×48 (D_f = 192)
deformable attn으로 3D→2D 투영 · 4 layers

축 방향 평균 → P_xy · P_xz · P_yz

Flex(2025, nvidia)



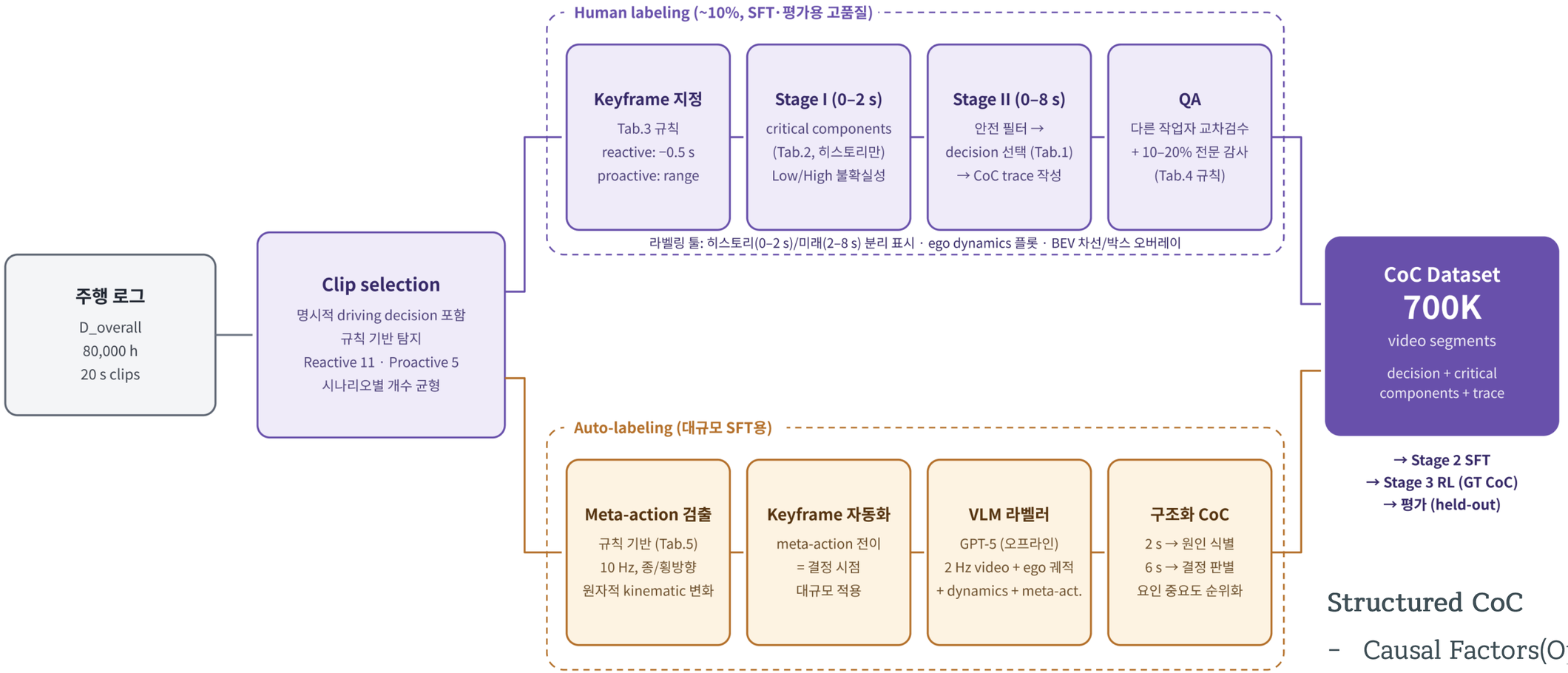
C cams × T timesteps 이미지 (2×9 = 18장)

DINOv2 patchifier
+ timestep PE (DiT) · camera-type embedding

[K scene tokens ; 모든 image tokens]
8-layer full self-attn (cross-view · cross-time)

image tokens 폐기 → K scene tokens (K = 900)

3. CoC Dataset



Structured CoC

- Causal Factors(Open-set)
- Driving Decision(Close-set)
- Composed CoC Traces

3. CoC Dataset

Table 1: Closed-set driving decisions (longitudinal and lateral) used to anchor reasoning traces to ex intent. Annotators select at most one decision per channel (or *None*), ensuring decision-grounded Definitions emphasize operational intent and disambiguate visually or behaviorally similar maneuvers (e.g., *Lead obstacle following* vs. *Yield*, *Lane change* vs. *Merge / Split*). Each selected decision must be causally supported by evidence from the observed history window. LC denotes lane change.

Type	Driving decision	Definition
Longitudinal	Set speed tracking	Maintain or reach a target speed when unconstrained; excludes follow/yield/stop logic.
	Lead obstacle following	Maintain a safe time gap to the lead entity (closest in-path entity moves in the same traffic flow); excludes geometry-based slowing, gap-matching, and yielding to non-lead entity.
	Speed adaptation (road events)	Adjust speed for roadway features (curves, grades, bumps, ramps, roundabouts, turns); independent of a lead.
	Gap-searching (for LC/merge/zipper)	Adjust speed to match the target stream or create a usable gap to support a planned lateral maneuver.
	Acceleration for passing/overtaking	Increase speed to pass a slower lead with an associated lateral plan.
	Yield (agent right-of-way)	Slow/stop to concede priority to specific agents (pedestrians, cross-traffic, emergency vehicles, cut-ins).
	Stop for static constraints	Decelerate to—and hold at—control points (stop/yield lines, red light, school bus/rail rules); Sometimes a yield is necessary even when owning the right-of-way, to avoid a collision.
Lateral	Lane keeping & centering	Maintain position within lane boundaries; minor in-lane offsets allowed; never cross lane lines.
	Merge / Split (facility change)	Transition between facilities (e.g., on-ramp ↔ mainline, weave segments); not a same-road lane change.
	Out-of-lane nudge (straddle avoidance)	Brief, intentional lane-line crossing to increase clearance around a blockage/hazard; return to original lane; specify left/right.
	In-lane nudge	Temporary offset within the lane (no line crossing) to increase clearance around a blockage/hazard; specify left-/right.
	Lane change (lateral push)	Full adjacent-lane transition with gap negotiation; specify left/right in reasoning trace.
	Pull-over / curb approach	Move toward edge/shoulder or a designated stop area (pickup, emergency stop, parking approach).
	Turn (intersection/roundabout/U-turn)	Planned path onto a different road segment with a significant heading change; specify left/right.
	Lateral maneuver abort	Cancel an ongoing lateral maneuver (nudge, lane change, merge/split, pull-over) and re-center when safe.

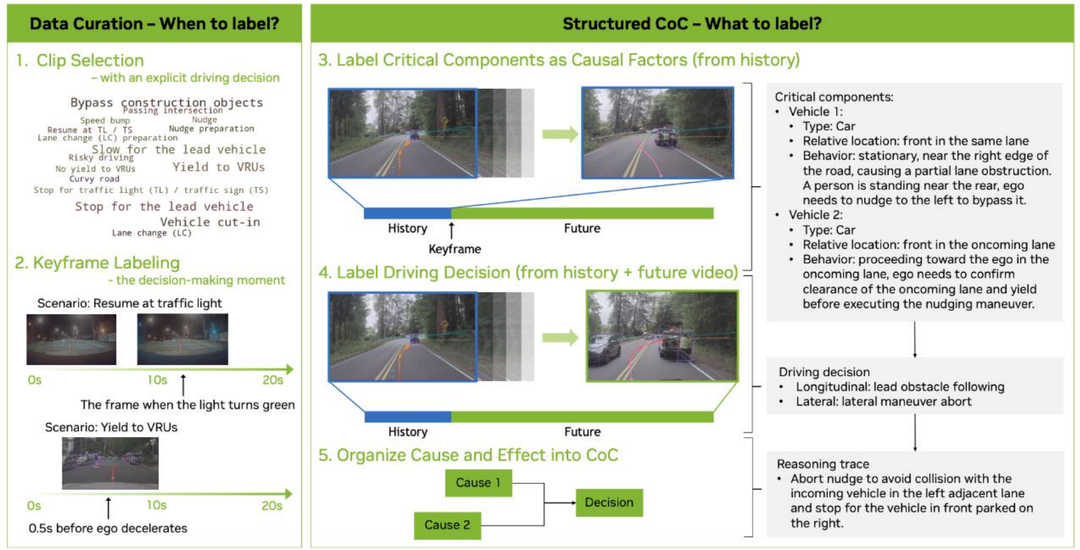
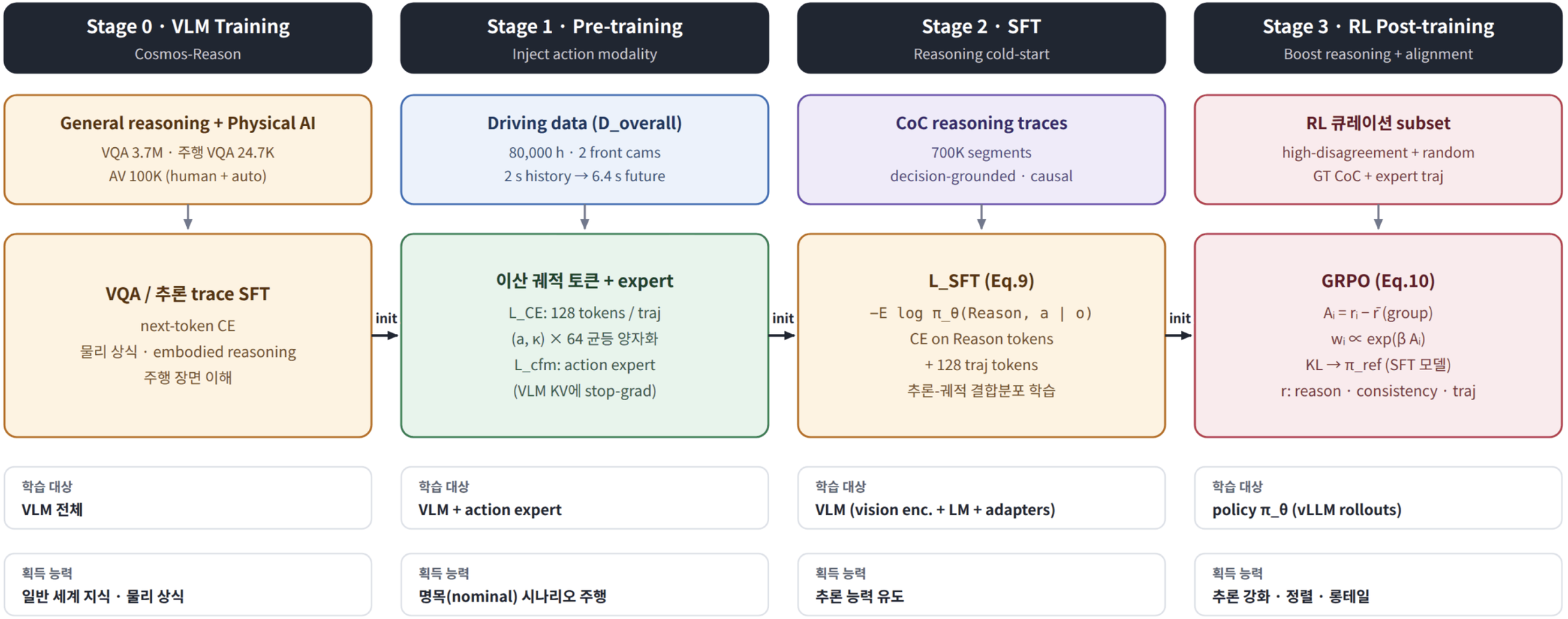


Figure 3: Overview of the proposed structured CoC labeling pipeline, composed of five steps: (1) **Clip Selection**, where clips containing explicit driving decisions are selected, filtering out low-signal clips that offer limited causal information; (2) **Keyframe Labeling**, where the decision-making moment within each video clip is identified, minimizing potential causal confusion; (3-5) **Structured CoC Labeling**, to construct the final CoC and further mitigate causal confusion, we first annotate critical components from the observation while avoiding referring to causal factors in future frames, and then label the corresponding driving decision. We then compose a reasoning trace from driving decisions and causal factors in natural language.

4. Training Pipeline



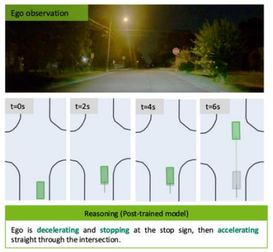
5. RL Post-Training

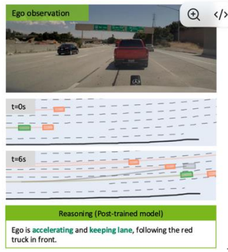
LRT critic





Reason Consistency





Scene o
멀티카메라
+ ego history
(+ route)

Policy π_θ
Alpamayo-R1
init: SFT 모델
(= π_ref)

sample K

rollout τ_i, i = 1…K
Reason_pred (CoC)
trajectory τ_pred
(평가 시 K = 6)

Reason

r_reason · LRM critic
DeepSeek-R1 / Cosmos-Reason (평가자)
입력: 히스토리 마지막 프레임 멀티캠 이미지
+ Reason_GT + Reason_pred
rubric 0–5: behavior 일치 · causal 일치
generation–verification gap 활용

r_consistency · CoC ↔ Action
τ_pred → meta-actions (규칙 기반)
Reason_pred → driving decision 파싱
종·횡 두 축 모두 일치 → 1, 아니면 0
결정 집합으로 파싱 실패 → 0 (보수적)

r_traj · 궤적 품질 (Eq.11)
$\lambda_L2\ \|x_pred - x_expert\|^2$
$+\lambda_coll \cdot \mathbb{1}[collision(x_pred)]$
$+\lambda_jerk \cdot J(x_pred)$

보상 결합
$r_i \leftarrow$ {r_reason, r_consistency, r_traj}

Group advantage
$A_i = r_i - \bar{r}$
$w_i = softmax(\beta A_i)$

GRPO 업데이트
$-E[\, w_i \,(\log \pi_\theta(\tau_i)$
$- \lambda_KL\ KL[\pi_\theta \,\|\, \pi_ref]\,)\,]$
π_ref = SFT 모델

6. Results & 생각해볼 것들

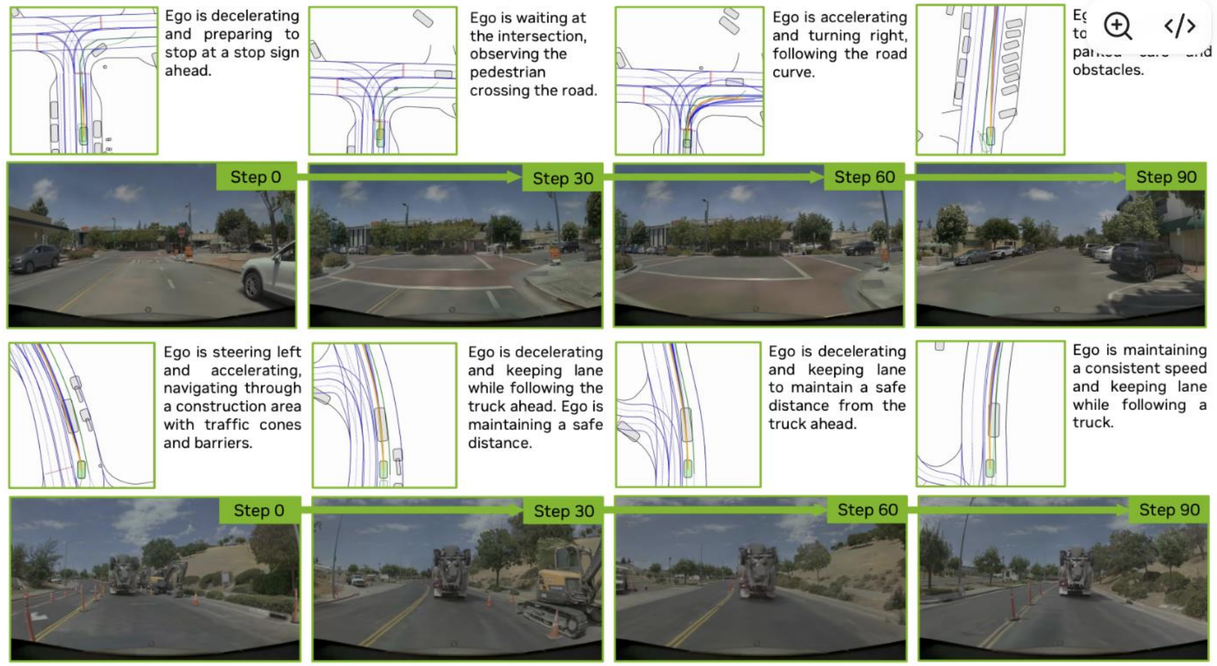


Figure 9: Examples of closed-loop evaluation in AlpaSim (NVIDIA et al., 2025). The top row presents an intersection scenario, whereas the bottom row illustrates a construction scenario.

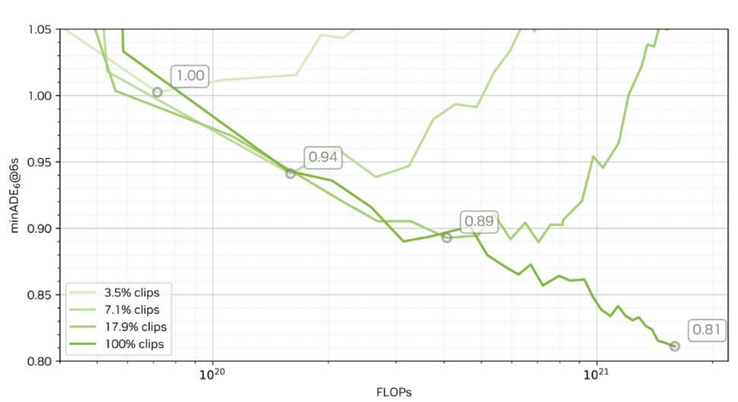


Figure 13: Impact of training data scale on open-loop driving performance. All models use the 0.5B architecture with identical hyperparameters and fixed total training steps. Models are evaluated on $\mathcal{D}_{\text{overall}}$ held-out test set.

Table 6: Open-loop evaluation of models on the CoC dataset. The base model is pre-trained with $\mathcal{D}_{\text{o}}$ other models are finetuned on the CoC dataset, then evaluated on held-out CoC test data. Numbers background are the best under each setting.

ID	Model Name	Route	Parameters	minADE_6@3s↓	minADE_6@6.4s↓
1	Base model (action modality)	×	0.5B	0.284	0.996
2	+ Ft. w/ Traj.	×	0.5B	0.282	0.971
3	+ Ft. w/ Meta-action & Traj.	×	0.5B	0.291	0.988
4	+ Ft. w/ CoC & Traj. (AR1)	×	0.5B	0.279	0.955
5	Base model (action modality)	×	3B	0.291	0.977
6	+ Ft. w/ Traj.	×	3B	0.293	0.976
7	+ Ft. w/ Meta-action & Traj.	×	3B	0.280	0.927
8	+ Ft. w/ CoC & Traj. (AR1)	×	3B	0.275	0.908
9	Base model (action modality)	✓	0.5B	0.264	0.848
10	+ Ft. w/ Traj.	✓	0.5B	0.262	0.834
11	+ Ft. w/ Meta-action & Traj.	✓	0.5B	0.264	0.821
12	+ Ft. w/ CoC & Traj. (AR1)	✓	0.5B	0.254	0.794

6. Results & 생각해볼 것들



- Vison Encoder 최적화
- Decoder 최적화
- 대규모 CoC가 필요할까?

THANK YOU
FOR YOUR ATTENTION

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